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| **Fiscal Year:** | FY 2010 | **Task Last Updated:** | FY 03/26/2010 |
| **PI Name:** | Mudawar, Issam Ph.D. | | |
| **Project Title:** | Ground-based measurement and mechanistic modeling of dryout in reduced-gravity flow boiling, and transitioning to tests on the International Space Station (ISS) | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | FLUID PHYSICS | | |
| **Program/Discipline--Element/Subdiscipline:** | FLUID PHYSICS--Fluid physics | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | West Lafayette | **State:** | IN |
| **Zip Code:** | 47907-1288 | **Congressional District:** | 4 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2008 Fluid Physics NNH08ZTT002N |
| **Start Date:** | 06/01/2009 | **End Date:** | 05/31/2012 |
| **No. of Post Docs:** | | **No. of PhD Degrees:** | |
| **No. of PhD Candidates:** | 1 | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA GRC |
| **Contact Monitor:** | | **Contact Phone:** | |
| **Contact Email:** | | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX09AJ51A | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |
| | Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future manned missions back to the Moon and afterwards to Mars. The emphasis in two-phase fluid physics is evident from recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling, dryout and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for all these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those | | |

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| **Task Description:** | three topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments that were conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer, dryout and condensation. Preparations for future intermediate parabolic flight experiments and long-term ISS experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | There are two overriding objectives for the proposed study:<br>(i) Develop an experimentally validated, mechanistic model for the minimum flow velocity required to ensure gravity independent flow boiling heat transfer and dryout.<br><br>(ii) Develop an experimentally validated, mechanistic model for the minimum flow velocity required to ensure gravity independent condensation heat transfer.<br><br>Specific tasks to meet these objectives can be summarized as follows:<br><br>Flow Boiling and Dryout Study:<br><br>(i) Modify an existing heat-flux-controlled Purdue University Parabolic Flight Flow Boiling CHF Apparatus to accommodate temperature-controlled flow boiling experiments. This task has been largely accomplished.<br><br>(ii) Modify this test facility to be used as a benchmark for a large-scale facility for micro-gravity experiments on the International Space Station (ISS). This task has been accomplished except for a condensation module to be incorporated into the flow loop.<br><br>(iii) Perform a parametric study of flow boiling and dryout of FC-72 in a test module that could be tilted to different orientations relative to Earth's gravity. This work will commence during the second year of the project.<br><br>(iv) Design a flow boiling and condensation module to integrate with a NASA Microgravity Science Glovebox (MSG) for experiments on the International Space Station (ISS). The flow boiling module for this facility has aleady been developed and successfully tested.<br><br>(v) Explore bubble coalescence, annular flow regime development and dryout using high-speed video and photomicrographic techniques. This task will commence during the second year of the project.<br><br>(vi) Develop a mechanistic model for the influence of flow velocity on flow boiling heat transfer and dryout in reduced gravity. This task will be completed during the third year of the project.<br><br>(vii) Develop dimensionless criteria for the minimum velocity required to negate the effects of gravity level on flow boiling heat transfer and dryout. This task will be completed during the third year of the project.<br><br>Condensation Study:<br><br>(i) Construct a large-scale test facility for the study of condensation heat trasnfer at different orientations relative to gravity. This task has already been largely accomplished.<br><br>(ii) Explore interfacial behavior of the condensation film using high-speed video and photomicrographic techniques. This task will commence during the second year of the project.<br><br>(iii) Perform condensation heat transfer measurements at different orientations relative to gravity. This task will commence during the second year of the project.<br><br>(iv) Develop a theoretical model for laminar and turbulent annular film condensation capable of predicting the influence of orientation relative to gravity. This task will be performed during the third year of the project. |
| **Bibliography Type:** | Description: (Last Updated: 01/06/2026) |
| **Articles in Peer-reviewed Journals** | Zhang H, Mudawar I, Hasan MM. "Application of flow boiling for thermal management of electronics in microgravity and reduced-gravity space systems." IEEE Transactions on Components and Packaging Technologies, 2009 Jun;32(2):466-77. http://dx.doi.org/10.1109/TCAPT.2008.2004413 , Jun-2009 |
| **Awards** | Lee J, Mudawar I. "One of most cited articles for the years 2005-2008, International Journal of Heat and Mass Transfer, for the paper. 'Two-Phase Flow in High-Heat-Flux Micro-Channel Heat Sink for Refrigeration Cooling Applications: Part I - Pressure Drop Characteristics,' Vol. 48, pp. 928-940, 2005." Dec-2009 |
| **Awards** | Mudawar I. "Rolls-Royce Milestone Award for design of air-to-fuel heat exchanger for high Mach aircraft turbine engines, LibertyWorks, North American Technologies operations of Rolls-Royce, September 2009." Sep-2009 |
| **Awards** | Lee J, Mudawar I. "Top 10 most download articles in the ASME Journal for Electronic Packaging for July and August 2009, for the paper, 'Experimental Investigation and Theoretical Model for Subcooled Flow Boiling Pressure Drop in Microchannel Heat Sinks,' Vol. 131, 031008, 2009." Sep-2009 |
| **Awards** | Sung MK, Mudawar I. "Top 10 most download articles in the ASME Journal for Electronic Packaging for June and July 2009, for the paper, 'Single-Phase and Two-Phase Hybrid Cooling Schemes for High-Heat-Flux Thermal Management of Defense Electronics,' Vol. 131, 021013, 2009." Sep-2009 |

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| **Awards** | Khanikar V, Mudawar I, Fisher T. "Top 25 hottest articles in the International Journal of Heat and Mass Transfer, for the paper, 'Effects of Carbon Nanotube Coating on Flow Boiling in a Micro-Channel,' Vol. 52, pp. 3805-3817, 2009." Dec-2009 |